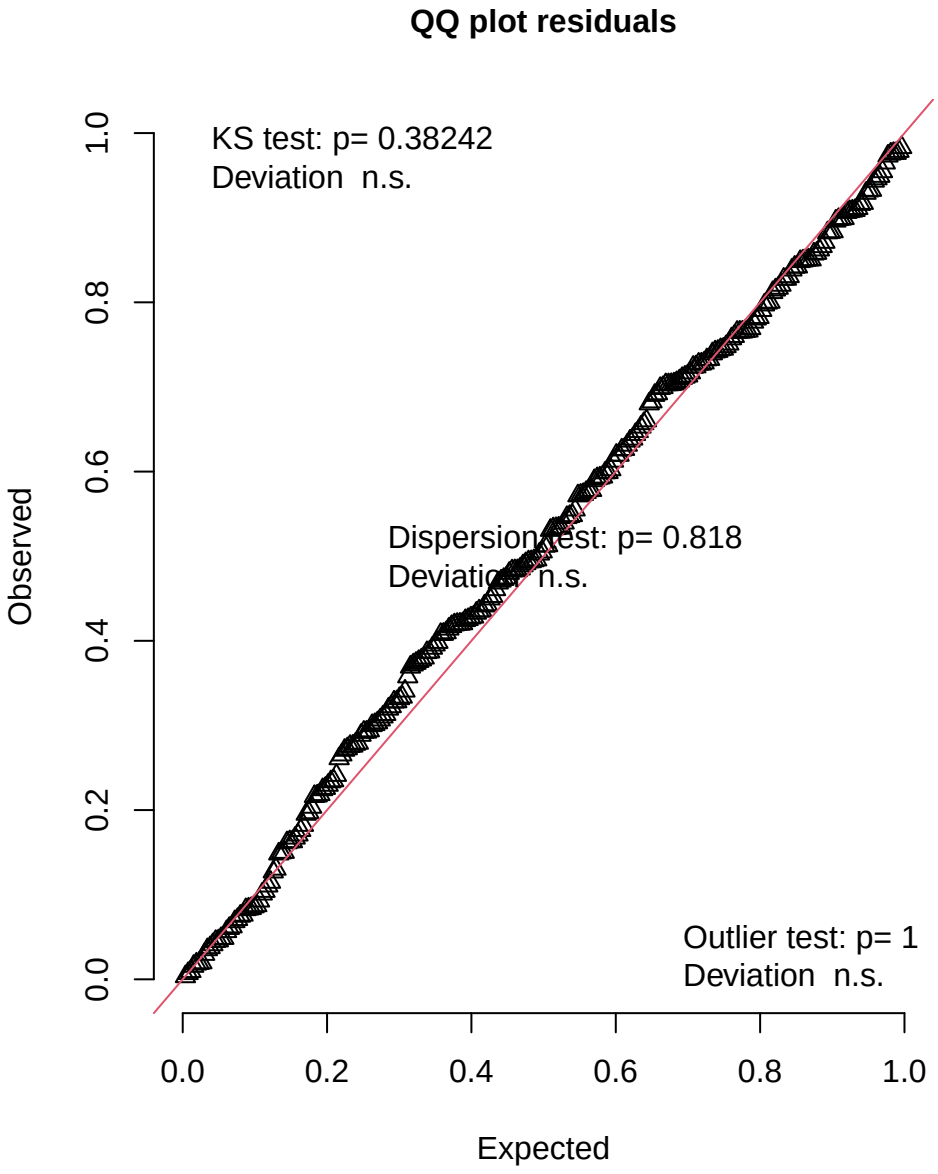
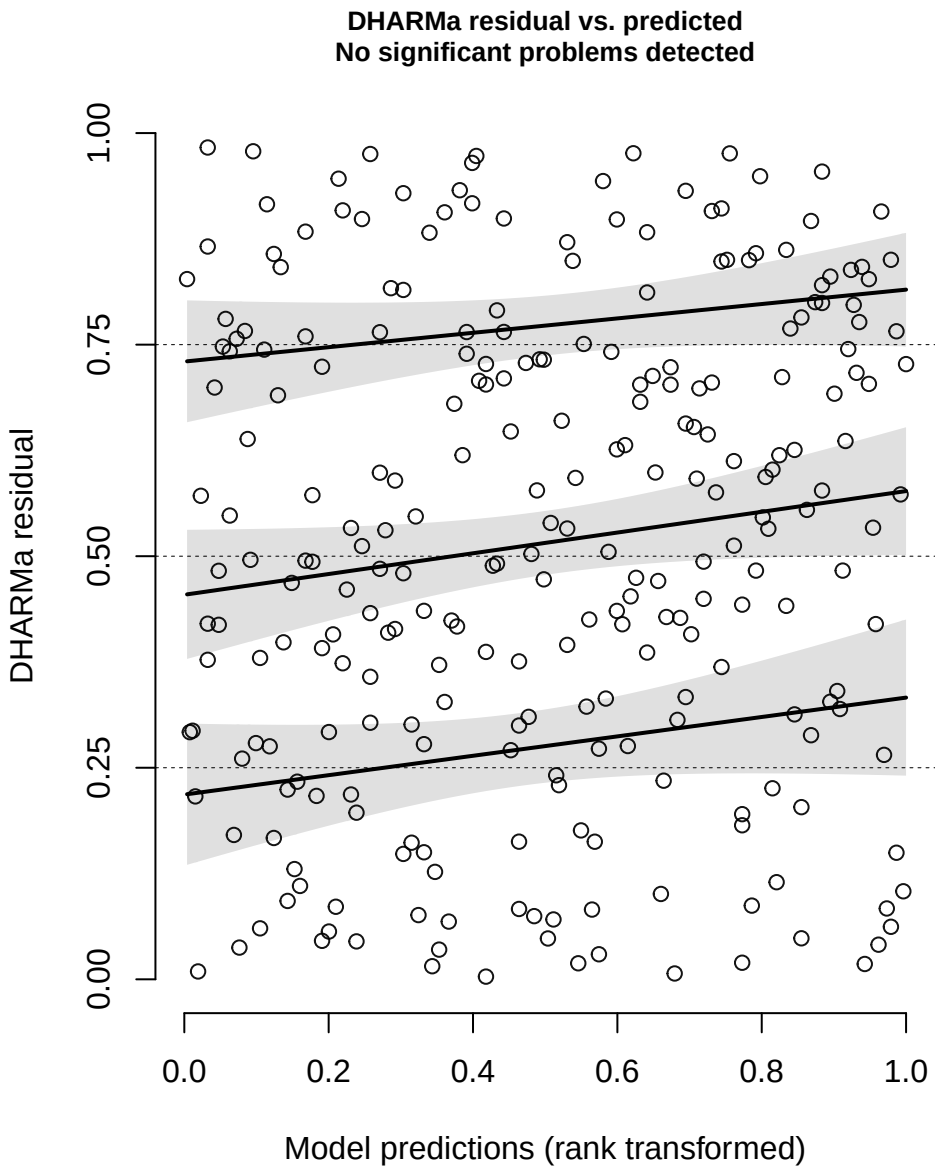
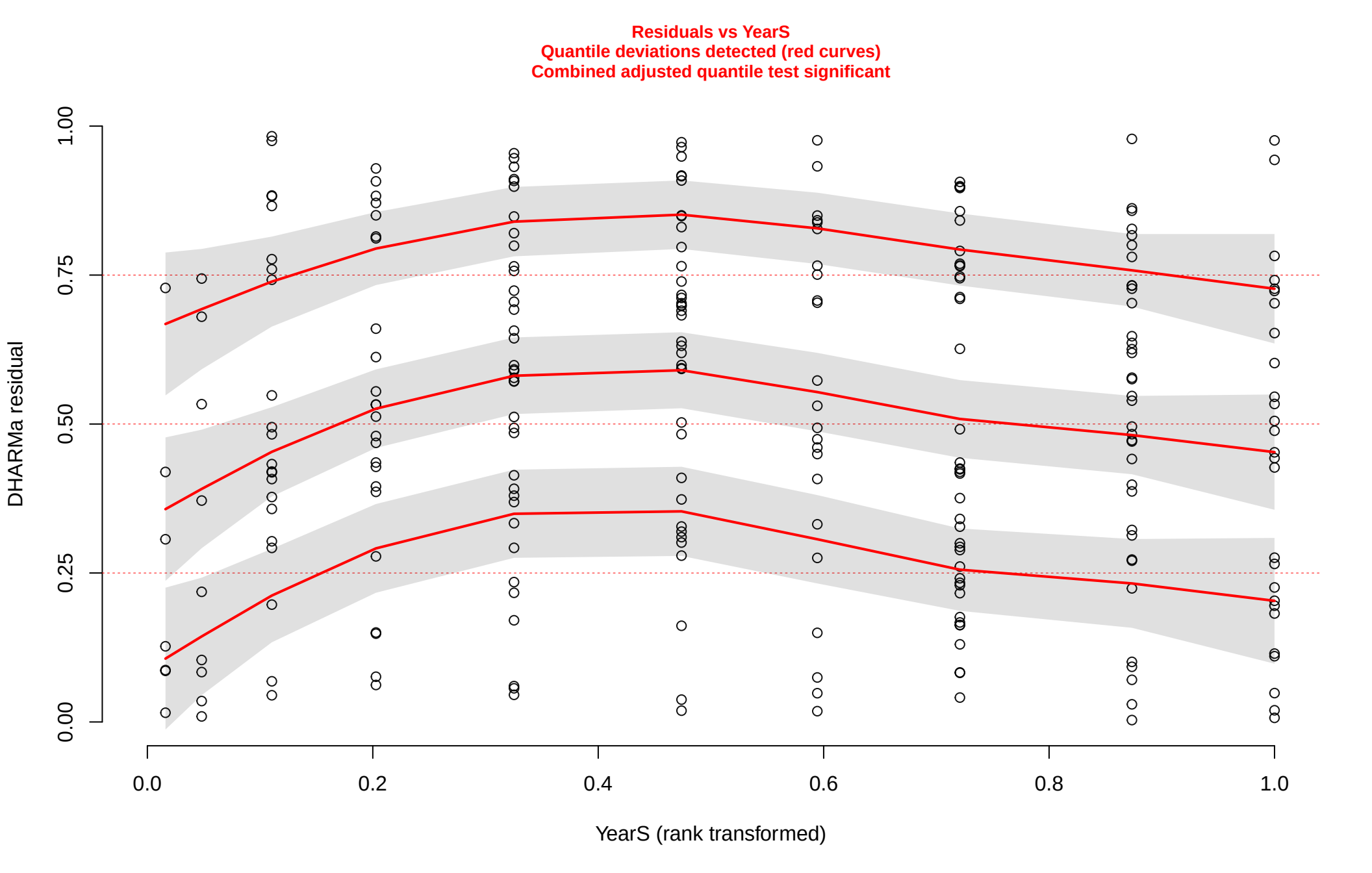
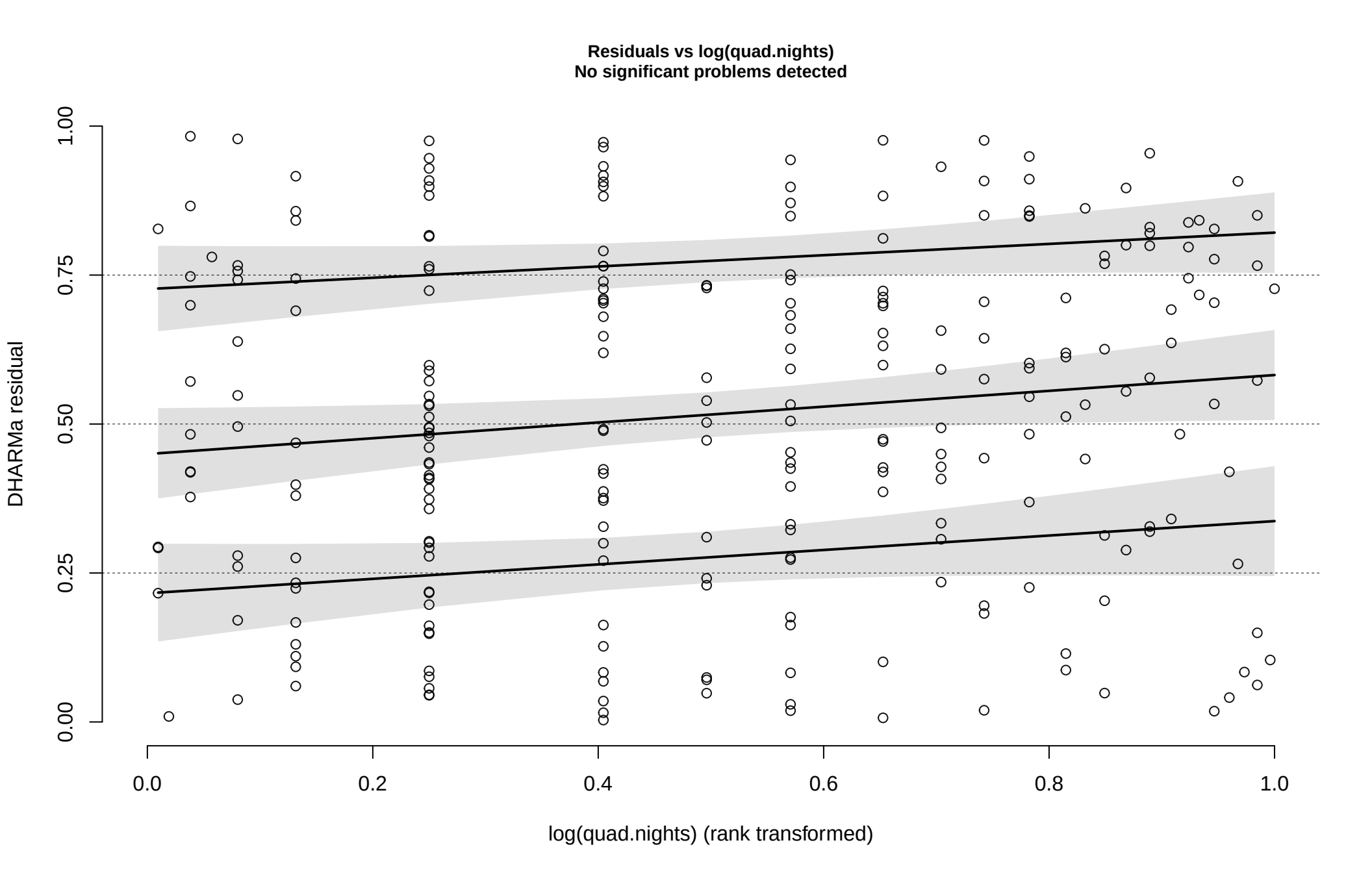
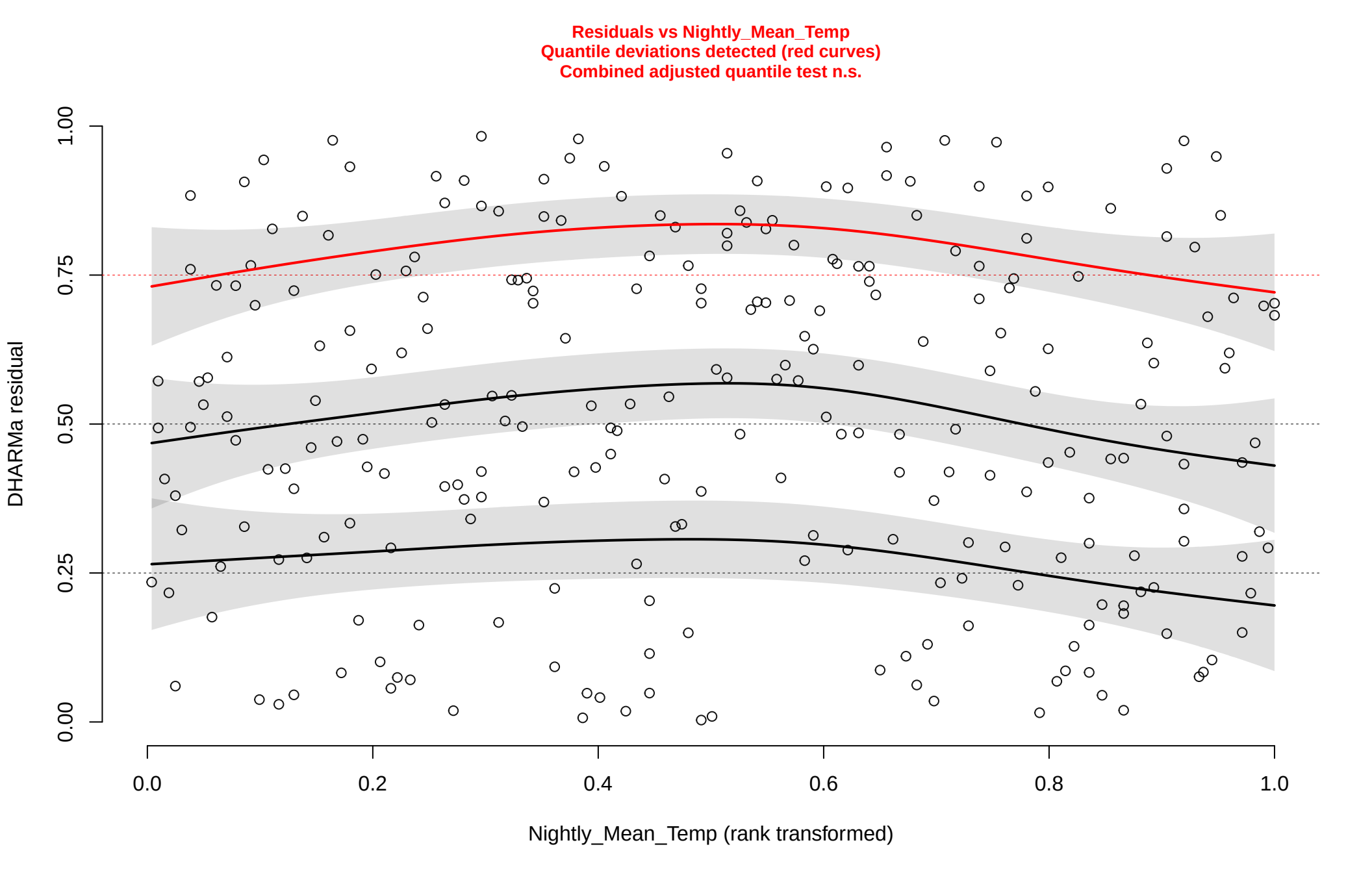
DHARMa residuals: EUMA

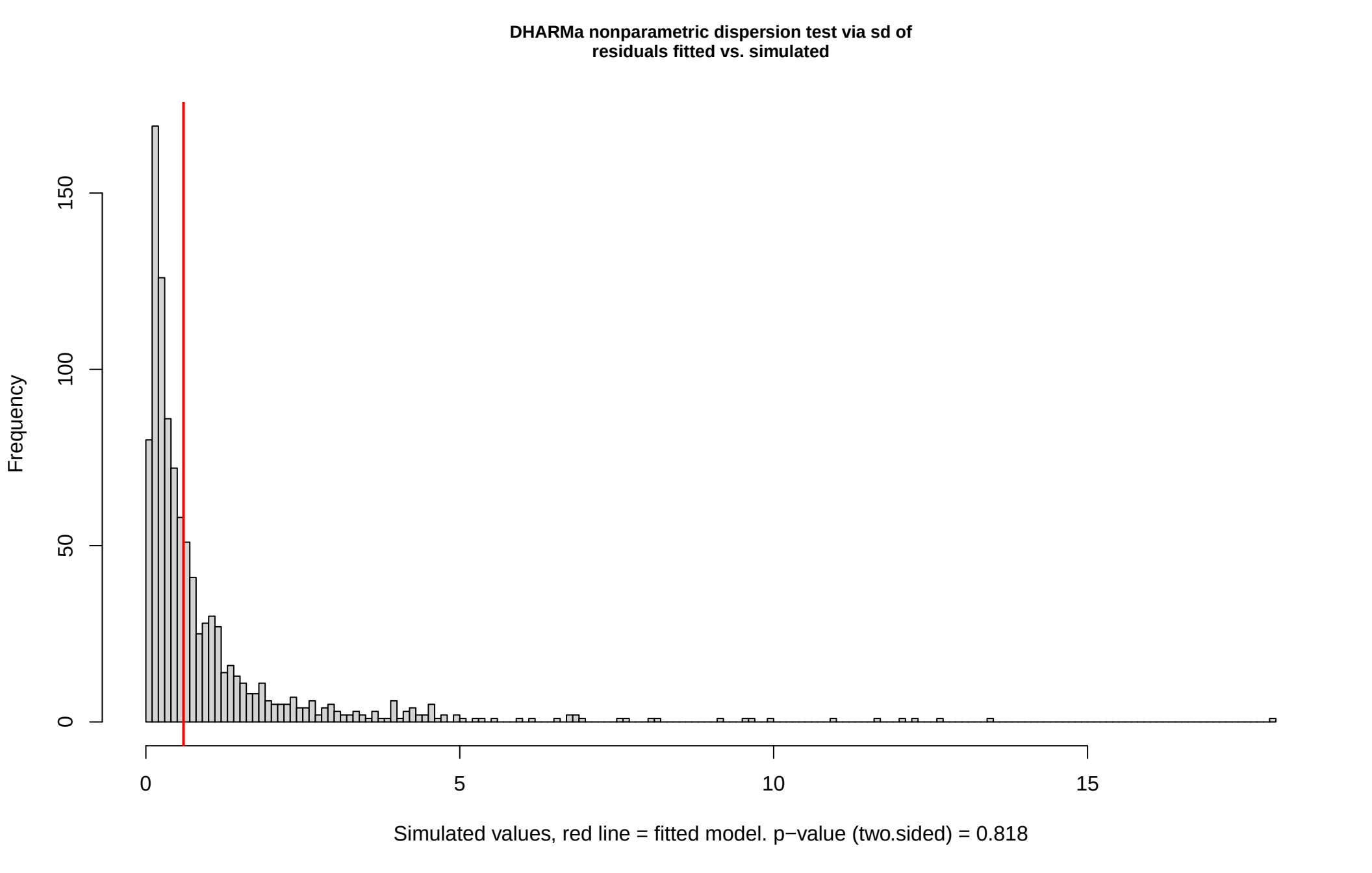


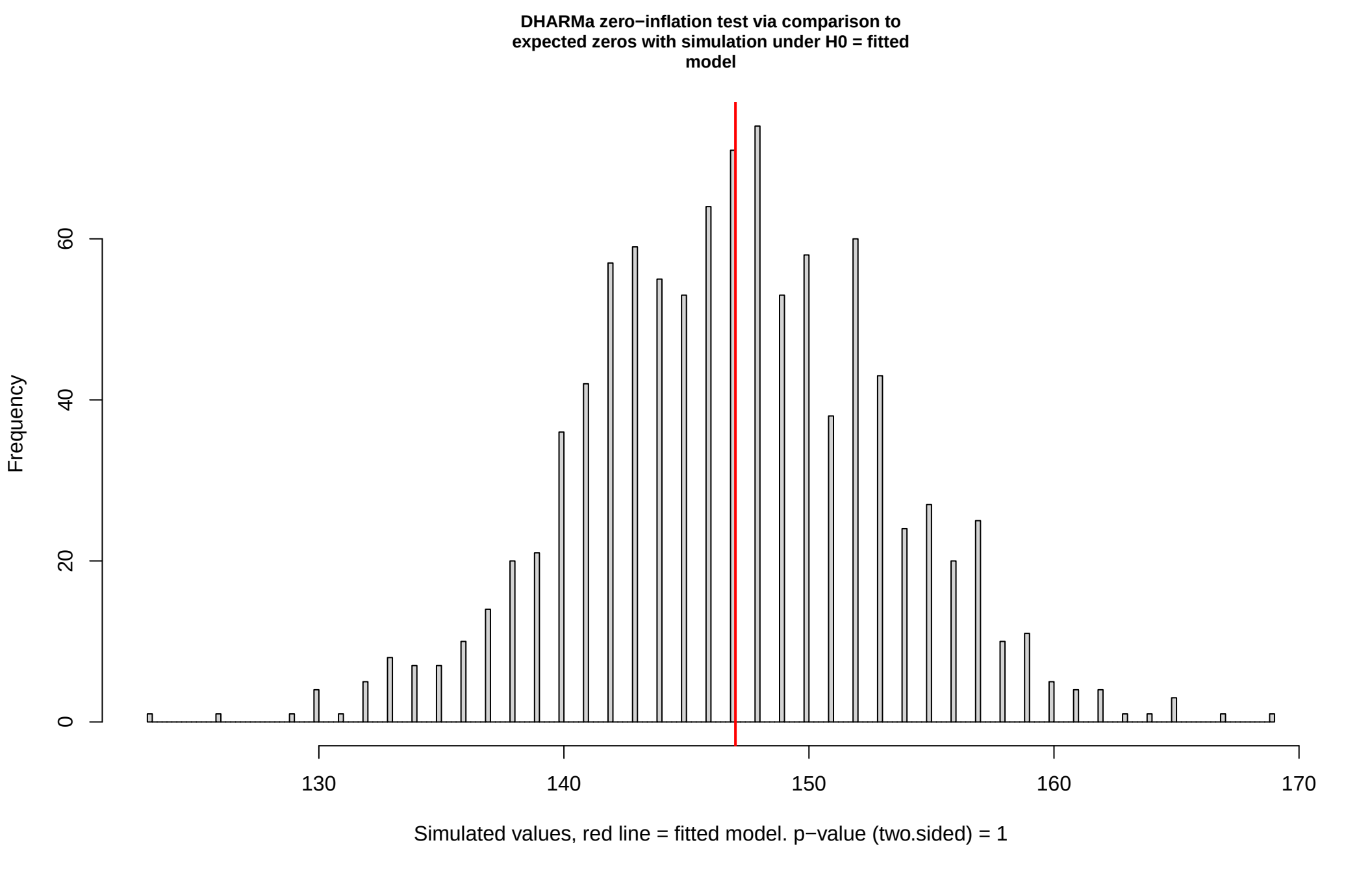



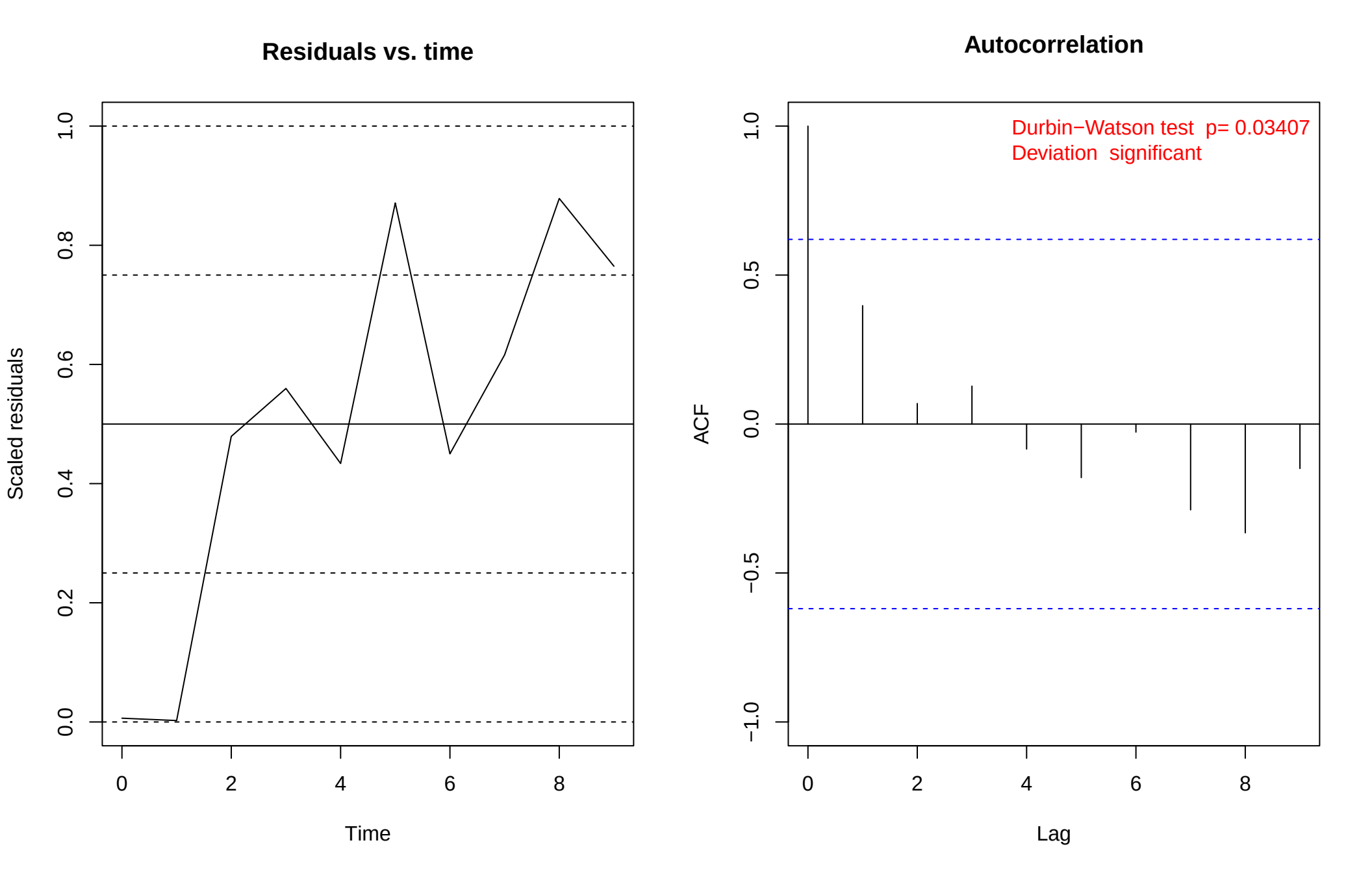

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

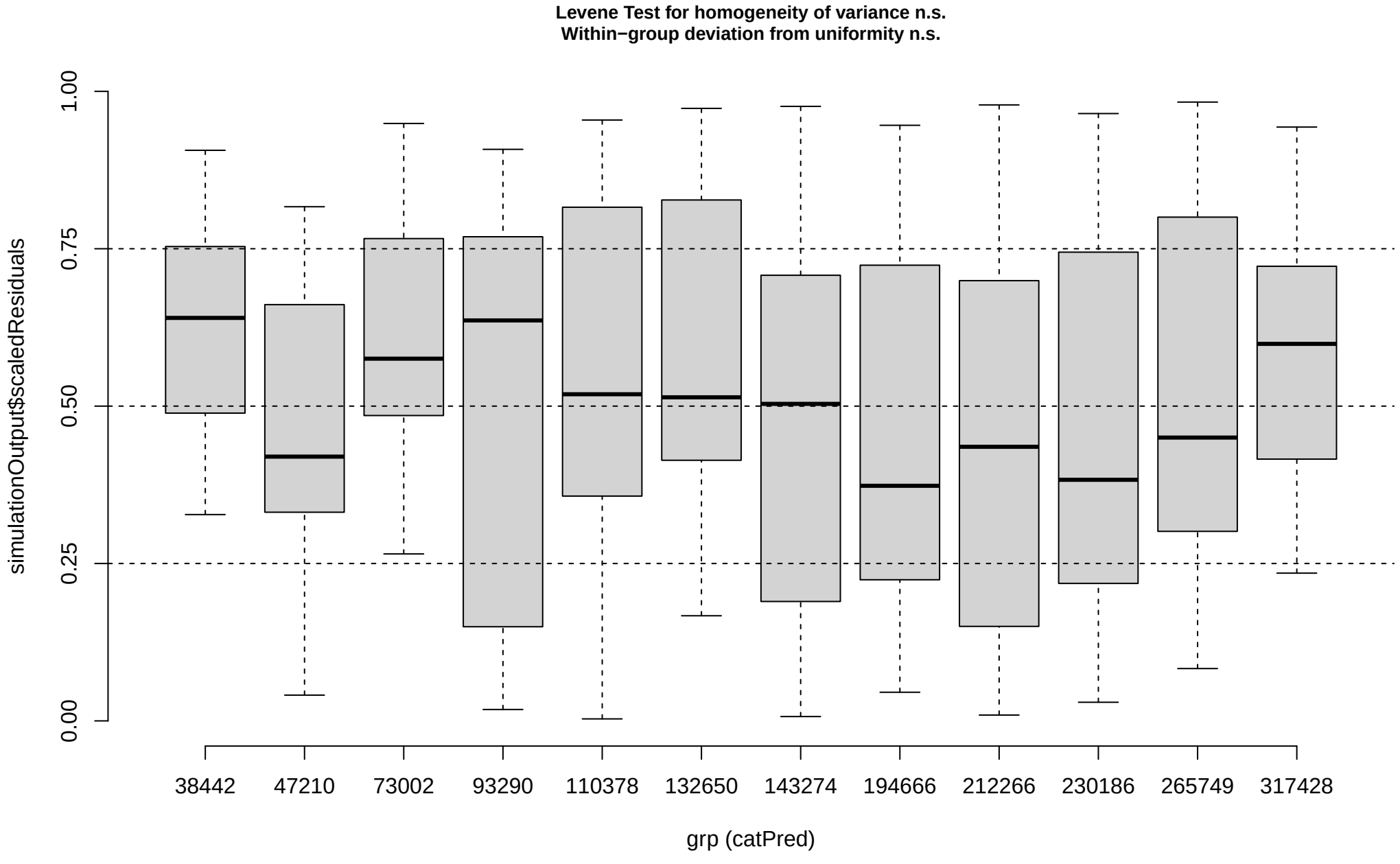

Residuals vs log(quad.nights)
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

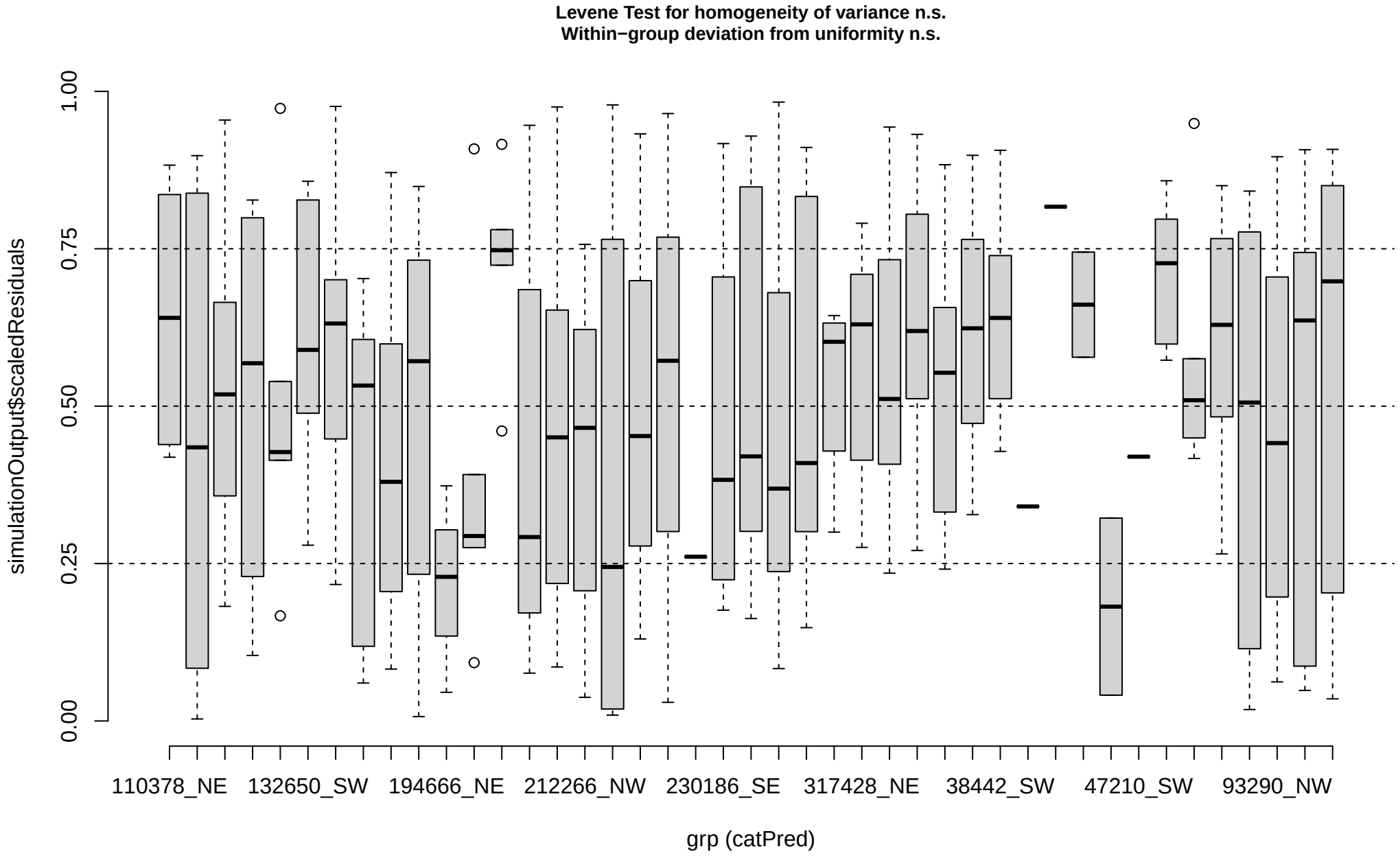

Residuals vs Nightly_Mean_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

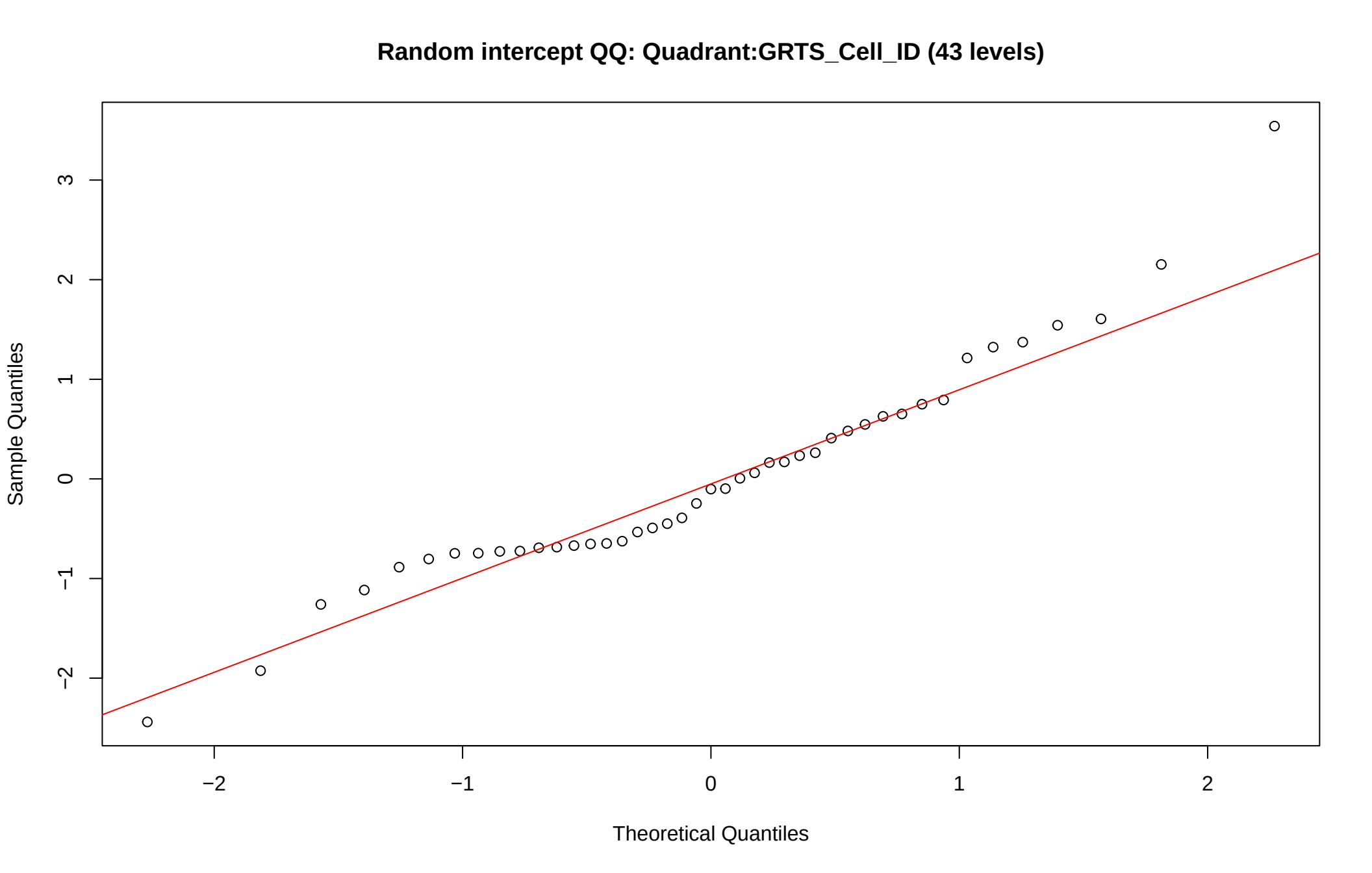

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
50
100
150
0
5
10
15
Simulated values, red line = fitted model. p−value (two.sided) = 0.818

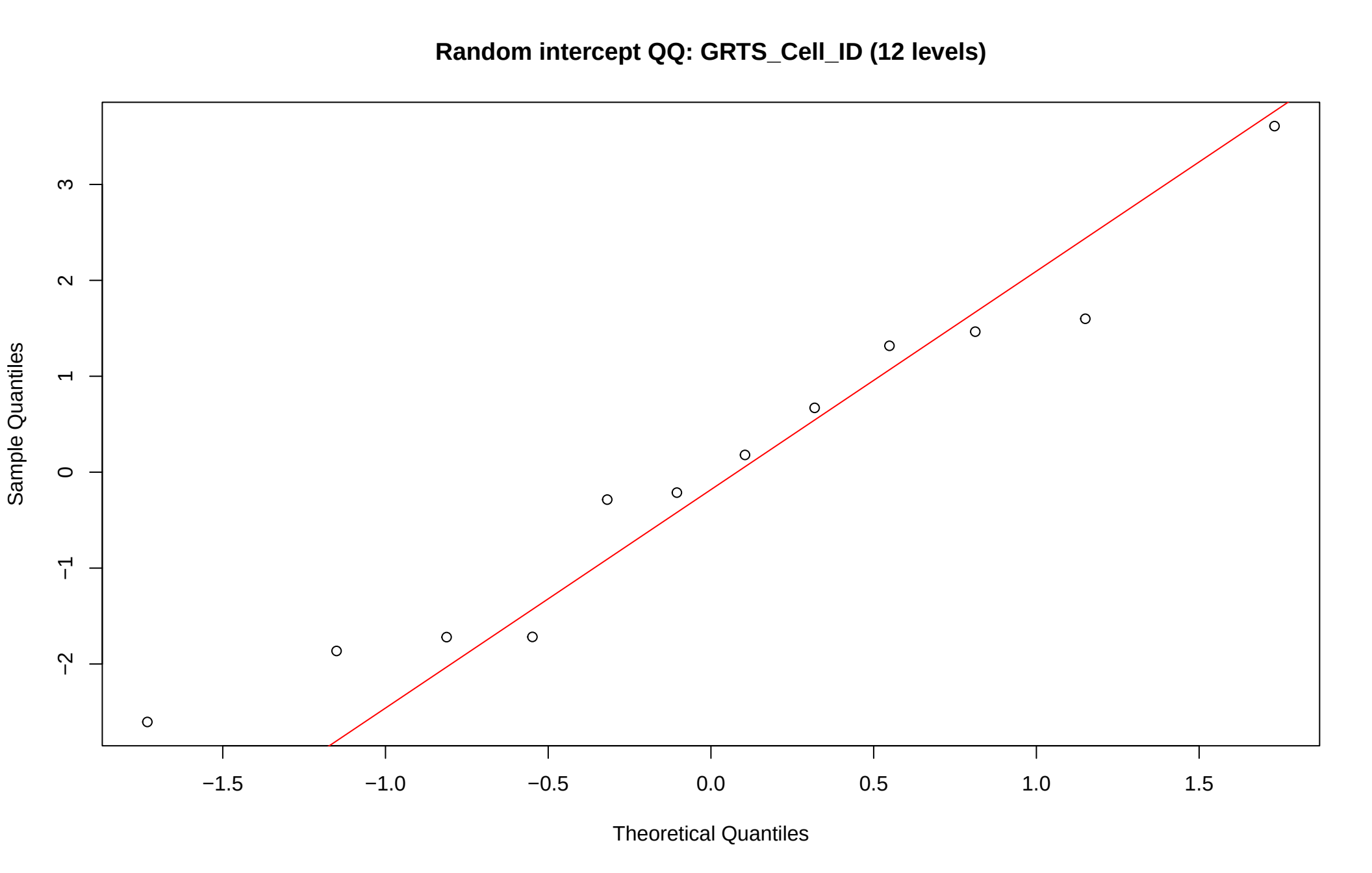

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
130
140
150
160
170
Simulated values, red line = fitted model. p−value (two.sided) = 1


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.03407
Deviation significant
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
38442
47210
73002
93290
110378
132650
143274
194666
212266
230186
265749
317428
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
110378_NE
132650_SW
194666_NE
212266_NW
230186_SE
317428_NE
38442_SW
47210_SW
93290_NW
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (43 levels)
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (12 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
3
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5